Hamiltonian Mechanics

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I. INTRODUCTION

Hamiltonian mechanics is a form of mechanics whose main advantage is being much more flexible when it comes to choosing a coordinate system than Newtonian or even Lagrangian mechanics. Similar to the Lagrangian ($\mathcal{L}$), Hamiltonian mechanics uses the Hamiltonian denoted by $\mathcal{H}$. However, for most systems, $\mathcal{H}$ ends up being equal to the total energy which gives it more meaning than $\mathcal{L}$.

II. EXAMPLE PROBLEM

Consider the following problem:

A 3kg frictionless block has a rocket strapped to it. The force of the rocket is given by $F = kx$ where $k = 10\text{N/m}$. In this paper we will learn how to model the motion of the rocket and compare trajectories from different initial conditions using the Hamiltonian.

III. DERIVING THE HAMILTONIAN

Before we derive the Hamiltonian, we need to define our coordinate system. The Hamiltonian is written in terms of generalized coordinates $q_1...q_i$ (for our example problem $q = x$) and also generalized momentum $p_i = \frac{\delta \mathcal{L}}{\delta \dot{q}_i}$. It was originally brought up in the *Classical Mechanics*[1] in chapter 7 in relation to the conservation of energy in Lagrangian mechanics. This is because the Hamiltonian is defined as an important quantity since it is conserved when $\frac{\delta \mathcal{L}}{\delta t} = 0$ (in other words, when the Lagrangian is not specifically dependent on t).

To derive this, we start with the Lagrangian in terms n number of generalized coordinates q_i and $\dot{q}_i$ and take the time derivative of it. Using the chain rule, we get:

$$\frac{d}{dt}\mathcal{L} = \sum_i \frac{\delta \mathcal{L}}{\delta q_i}\dot{q}_i + \sum_i \frac{\delta \mathcal{L}}{\delta \dot{q}_i}\ddot{q}_i + \frac{\delta \mathcal{L}}{\delta t}$$

Next, by Lagrange's equation and the generalized momentum defined earlier, we know:

$$\frac{\delta \mathcal{L}}{\delta q_i} = \frac{d}{dt}\frac{\delta \mathcal{L}}{\delta \dot{q}_i} = \frac{d}{dt}p_i = \dot{p}_i$$

So $\dot{p}_i$ can replace the derivative in the first sum and the derivative in the second sum is p so we can rewrite the time derivative of the Lagrangian as:

$$\frac{d}{dt}\mathcal{L} = \sum_i (\dot{p}_i\dot{q}_i + p_i\ddot{q}_i) + \frac{\delta \mathcal{L}}{\delta t}$$

$$= \frac{d}{dt}\sum_i (p_i\dot{q}_i) + \frac{\delta \mathcal{L}}{\delta t}$$

When the Lagrangian does not explicitly depend on time, meaning $\frac{\delta \mathcal{L}}{\delta t} = 0$, then the time derivative of $\sum p_i\dot{q}_i - \mathcal{L}$ is zero. This second expression is defined as the Hamiltonian and is conserved whenever the Lagrangian expression does not include t.

$$\mathcal{H} = \sum_{i=1}^{n} p_i\dot{q}_i - \mathcal{L}$$

While $\dot{q}$ is in the definition of the Hamiltonian, having both p and $\dot{q}$ is redundant since p is a function of $\dot{q}$. Therefore, the Hamiltonian is always written in terms of just p and q. It is also interesting to note that if the relation between Cartesian coordinates and the generalized coordinates is time independent, then the Hamiltonian is the total energy: $\mathcal{H} = T + U$. This is something that will be shown for 1 dimension in the next section, but a more general proof can be found in section 7.8 of *Classical Mechanics*[1].

IV. HAMILTONIAN IN 1 DIMENSION

Since our example problem is 1 dimensional, there is only one q and one corresponding p which eliminates the sum in the Hamiltonian and shortens it to:

$$\mathcal{H} = p\dot{q} - \mathcal{L}$$

We also will use the result from section 7.8[1] that using natural coordinates (meaning nothing explicitly in terms of t) in a conservative system results in the following general equation for the Lagrangian.

$$\mathcal{L} = \frac{1}{2}A(q)\dot{q}^2 - U(q)$$

Then, the generalized momentum is

$$p = \frac{\delta\mathcal{L}}{\delta\dot{q}} = A(q)\dot{q}$$

and from this we can see that $p\dot{q} = A(q)\dot{q}^2 = 2T$ so we can substitute this into the Hamiltonian to get

$$\mathcal{H} = p\dot{q} - \mathcal{L} = 2T - (T - U) = T + U$$

which is the result mentioned in the previous section that for many cases (including the example problem) $\mathcal{H}$ is the total energy of the system. For the next step we need to define the Hamiltonian in terms of just q and p. To do this we just rearrange the previous equation for generalized momentum ($p = A(q)\dot{q}$) to get

$$\dot{q} = \frac{p}{A(q)} = \dot{q}(q, p)$$

Now, we can use this to rewrite the Hamiltonian:

$$\mathcal{H} = p\dot{q}(q, p) - \mathcal{L}(q, \dot{q}(q, p))$$

Finally, we get our two equations of motion by taking the derivative of the above equation with respect to p and q. First, differentiating with respect to q, gives

$$\frac{\delta\mathcal{H}}{\delta q} = p\frac{\delta\dot{q}}{\delta q} - \left[\frac{\delta\mathcal{L}}{\delta q} + \frac{\delta\mathcal{L}}{\delta\dot{q}}\frac{\delta\dot{q}}{\delta q}\right]$$

Remember that $\frac{\delta\mathcal{L}}{\delta\dot{q}} = p$ so the first and last terms on the right of the equation cancel out leaving

$$\frac{\delta\mathcal{H}}{\delta q} = -\frac{\delta\mathcal{L}}{\delta q} = -\frac{d}{dt}\frac{\delta\mathcal{L}}{\delta\dot{q}} = -\frac{d}{dt}p = -\dot{p}$$

Next, taking the the derivative with respect to p and substituting in p for $frac\delta\mathcal{L}\delta\dot{q}$ again we get

$$\frac{\delta\mathcal{H}}{\delta p} = \left[\dot{q} + p\frac{\delta\dot{q}}{\delta p}\right] - \frac{\delta\mathcal{L}}{\delta\dot{q}}\frac{\delta\dot{q}}{\delta p} = \dot{q}$$

We now have our two Hamiltonian equations of motion:

$$\frac{\delta\mathcal{H}}{\delta q} = -\dot{p}$$

and

$$\frac{\delta\mathcal{H}}{\delta p} = \dot{q}$$

V. USING THE HAMILTONIAN

First we need to find the Lagrangian. For that we need kinetic energy T and potential energy U. We know

$$T = \frac{1}{2}m\dot{x}^2$$

and

$$U = -\int_0^x kx\,\mathrm{d}x = -\frac{1}{2}kx^2$$

. For this problem, it makes the most sense that we define are generalized coordinate as $q = x$. So that $T = \frac{1}{2}m\dot{q}^2$ and $U = -\frac{1}{2}kq^2$ So, the Lagrangian then is:

$$\mathcal{L} = T - U = \frac{1}{2}m\dot{q}^2 + \frac{1}{2}kq^2$$

and from that we can get the generalized momentum:

$$p = \frac{\delta\mathcal{L}}{\delta\dot{q}} = m\dot{q}$$

Remember that $q = x$, so p is just the momentum.
Now we can find the Hamiltonian:

$$\mathcal{H} = p\dot{q} - \mathcal{L} = \frac{1}{2}p\dot{q} - \frac{1}{2}kq^2$$

Writing in terms of p and q gives:

$$\mathcal{H} = \frac{p^2}{2m} - \frac{1}{2}kq^2$$

Now, we can substitute this into the two Hamiltonian equations of motion found at the end of the previous section. This gives us two coupled equations:

$$\dot{q} = \frac{\delta\mathcal{H}}{\delta p} = \frac{p}{m}$$

and

$$\dot{p} = -\frac{\delta\mathcal{H}}{\delta q} = kq$$

Since we finally have equations of motion, we can use an iterative method to graph q vs p for different initial conditions.

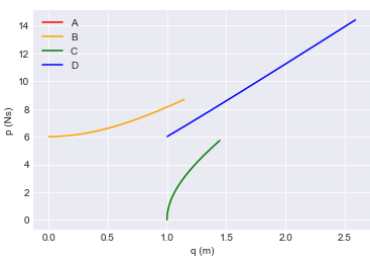

FIG. 1: This is a plot of p vs q from $t = 0$ to $t = 0.5$s for four different initial conditions. The initial conditions for each line are as follows:
A:$x_0 = 0$m,$p_0 = 0$Ns
B:$x_0 = 0$m,$p_0 = 6$Ns
C:$x_0 = 1$m,$p_0 = 0$Ns
D:$x_0 = 1$m,$p_0 = 6$Ns
The line for A does not show up because the rocket has no velocity at $t = 0$ and exerts no force at $x = 0$ so it just stays at the origin.

VI. LIOUVILLE'S THEOREM

Hamiltonian mechanics is often used (like in the graph above) to describe a set of identical systems with different initial conditions. In our graph, we only look at four initial conditions, but any number could be used and this is especially useful because Liouville's theorem allows us to model a cluster of identical systems as an uncompressible fluid.

Liouville's theorem states that the volume required to enclose a set of identical systems plotted with coordinates (q, p) is time-independent. This theorem is much easier to see in a graph so the following two figures which explain this theorem a bit better. Also, since we are working in only one dimension, there is only one p and one q making the volume from Liouville's theroem an area in our example. You can image though, that if our rocket moved in more dimensions, we would have more dimensions in our graph and so our points would fill a volume.

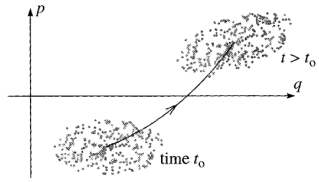

FIG. 2: This shows what Liouville's theorem means by a set or cluster of identical systems moving over time. An analogous graph for our problem would be Figure 1 if we plotted only the points at $t = 0$ and $t = 0.5$. This graph allows us to see how a cluster of identical systems moves over time in phase space. [1]

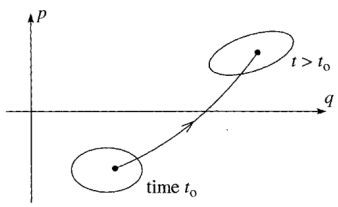

FIG. 3: This next graph is essentially the same as the previous one except it shows the "volume" (in this case it is an area) taken up by the cluster at two different times. What Liouvilles theorem states is that these to ellipses have the exact same area, and they will have the same area for any time t. [1]

The above graph also helps show the usefulness of Liouville's theorem. Instead of tracking many separate points, like in figure 2, it is possible to draw an area at $t = 0$ and then model how the area moves over time.

We can check our graph from the previous section (figure 1) by seeing if it follows Liouville's theorem. To do this, we will compare the areas outlined by the coordinates at $t = 0$ and $t = 0.5$.

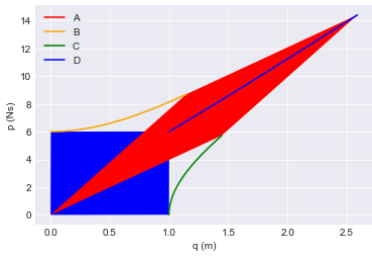

FIG. 4: Here we have the same graph as figure 1, but the two areas being compared are shaded. [1]

If you calculate the areas of the two shapes in figure 4, they both are 6 kg m/s^2 as predicted by Liouville's theorem. It is also interesting to note that if any one of the initial conditions is removed so that there are only 3 lines, both areas will be cut exactly in half and still be equal to each other which again follows the theorem.

VII. CONCLUSION

Finally, it is important to note that the only requirement for Liouville's theorem is that the systems are plotted in (q, p) coordinates. It does not matter if the Hamiltonian is conserved or if natural coordinates are being used, Liouville's theorem still applies. This theorem is one example of a major advantage Hamiltonian mechanics has over Lagrangian mechanics.

[1] J. R. Taylor, *Classical Mechanics* (University Science Books, 2005).